

W1437 Production Deployment Runbook

> **Release:** W1437 (feat/w1406-preflight-followup) > **PR:** [#579](#) > **Hotfix:** W1444 migration native-driver fix
> **Last updated:** 2026-06-22 > **Owner:** Release team / DevOps / on-call engineer

1. Purpose

This runbook describes the end-to-end procedure for deploying the W1437 release to production. W1437 fixes critical production DB timeouts and adds structured LLM anomaly diagnostics.

Mandatory pre-deploy step: run the `NphiesClaim updatedAt` backfill migration **before** deploying the new application code. Old `PENDING_REVIEW` claims will not be swept without it.

2. Prerequisites

Before starting, ensure:

- ☐ PR [#579](#) is merged to main.
 - ☐ Local main is up to date with origin/main.
 - ☐ Pre-push gates pass on main.
 - ☐ You have production SSH access (for VPS methods).
 - ☐ `MONGODB_URI` for the production cluster is available.
 - ☐ A MongoDB backup exists from the last 24 hours.
 - ☐ On-call engineer is notified.
 - ☐ Rollback plan is understood.
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3. Quick Reference

Artifact	Path
Migration script	<code>backend/scripts/migrate-nphies-claim-updatedAt.js</code>
VPS deploy executor	<code>scripts/deploy-w1437.sh</code>
VPS deploy integration	<code>scripts/deploy-vps.sh --with-w1437-migration</code>
Docker deploy	<code>scripts/deploy-w1437-docker.sh</code>
Rollback script	<code>scripts/rollback-w1437.sh</code>

Smoke tests	scripts/smoke-test-w1437.sh
VPS monitor	scripts/monitor-w1437.sh
Docker monitor	scripts/monitor-w1437-docker.sh
Loki alerts	ops/loki-rules-w1437.yml
GitHub migration workflow	.github/workflows/w1437-migrate.yml
GitHub monitor workflow	.github/workflows/w1437-monitor.yml
GitHub final-review workflow	.github/workflows/w1437-final-review.yml
Deployment status	docs/W1437_DEPLOY_STATUS.md
Deployment notes	docs/DEPLOYMENT_NOTES_W1437.md

4. Deployment Methods

Run the final review first:

```
./scripts/final-review-w1437.sh
```

Then choose **one** deployment method.

4.1 VPS / Bare-Metal (Recommended)

SSH to the production host and run:

```
cd /home/alawael/app # or your DEPLOY_ROOT
export MONGODB_URI="mongodb+srv://..."
export NODE_ENV=production
./scripts/deploy-w1437.sh --force
```

The script will: 1. Verify backup age and MongoDB connectivity. 2. Pull latest main. 3. Run the migration. 4. Verify compound indexes.

Then deploy backend + frontend as usual and run smoke tests:

```
./scripts/smoke-test-w1437.sh https://alaweal.org
```

Or with automatic rollback if smoke tests fail:

```
./scripts/smoke-test-w1437.sh --auto-rollback https://alaweal.org
```

Finally, start the 30-minute monitor:

```
./scripts/monitor-w1437.sh logs/error1.log
```

4.2 One-Command VPS Deploy

If you use the existing `deploy-vps.sh` script:

```
export MONGODB_URI="mongodb+srv://..."
./scripts/deploy-vps.sh --with-w1437-migration
./scripts/smoke-test-w1437.sh https://alaweal.org
./scripts/monitor-w1437.sh logs/error1.log
```

4.3 Docker Compose

```
export MONGODB_URI="mongodb+srv://..."
export COMPOSE_FILE="docker-compose.professional.yml:docker-compose.production.yml"
export COMPOSE_PROJECT_NAME=alawael
./scripts/deploy-w1437-docker.sh up
./scripts/smoke-test-w1437.sh https://alaweal.org
./scripts/monitor-w1437-docker.sh alawael-backend-1
```

4.4 Canary Deployment

Deploy to a canary host first, verify, then promote to production:

```
export MONGODB_URI="mongodb+srv://..."
export CANARY_HOST="canary.alaweal.org"
export PROD_HOST="alaweal.org"
export VPS_USER="..."
export VPS_SSH_KEY="$HOME/.ssh/deploy_key"
./scripts/deploy-canary-w1437.sh
```

4.5 Blue-Green Deployment

For zero-downtime cutover with instant rollback:

```
export MONGODB_URI="mongodb+srv://..."
export DEPLOY_ROOT=/home/alawael/app
export NGINX_COLOR_CONFIG=/etc/nginx/conf.d/alawael-color.conf
export SMOKE_URL_INACTIVE=http://localhost:3002
export SMOKE_URL_ACTIVE=https://alaweal.org
./scripts/deploy-bluegreen-w1437.sh
```

Requires two app directories (app-blue, app-green) and an nginx upstream config.

4.6 GitHub Actions

1. Go to **Actions** → **W1437 Production Migration** → **Run workflow**.
2. Type `migrate` in the confirm field.
3. Check **"Deploy to production"** to also deploy.
4. After deploy, run **Actions** → **W1437 Post-Deploy Monitor**.

Required secrets: `VPS_HOST`, `VPS_USER`, `VPS_SSH_KEY`, `MONGODB_URI`.

5. Dry Run

Estimate impact without writing anything:

```
DRY_RUN=1 NODE_ENV=production node backend/scripts/migrate-nphies-claim-updatedAt.js
```

Expected output:

```
[migrate-nphies-claim-updatedAt] dry-run { wouldMatch: 1234, wouldModify: 1234 }
```

6. Post-Deploy Monitoring

Watch `error1.log` (or container logs) for **30 minutes**. These signatures should **stop** after deploy:

Signature	Severity	Ticket
Operation advancedtickets.find() buffering timed out after 10000ms	P0	P0-1
Operation nphiesclaims.find() buffering timed out after 10000ms	P0	P0-2
[llm-anomaly-history] save failed:	P2	P2-4

Also verify:

- [] NPHIES reconciliation sweeper transitions old PENDING_REVIEW claims.
- [] SLA scheduler (`checkResolutionBreaches`, `assignMissingSlaDeadlines`, `getSlaStats`) completes without timeout errors.

7. Rollback

If the release causes problems, run:

```
export DEPLOY_ROOT=/opt/alawael-erp  
./scripts/rollback-w1437.sh
```

The script: - Backs up the current `backend/` and `frontend/` directories. - Resets the application code to the commit before W1437. - Restarts services. - **Does not** remove migration data (`nphies.submission.updatedAt`); old code ignores it safely.

To roll back to a specific commit or tag:

```
./scripts/rollback-w1437.sh v1.2.3
```

8. Alerts

8.1 Loki Log Alerts

Mount ops/loki-rules-w1437.yml into the Loki container:

```
volumes:  
- ./ops/loki-rules-w1437.yml:/loki/rules/fake/w1437.yml:ro
```

Alerts route to AlertManager at `http://alertmanager:9093`.

8.2 Notification Webhooks

The deploy scripts can notify Slack or Teams:

```
export SLACK_WEBHOOK="https://hooks.slack.com/services/..."  
# or  
export TEAMS_WEBHOOK="https://outlook.office.com/webhook/..."  
./scripts/deploy-w1437.sh
```

9. Feature Flags

W1437 behavior can be disabled at runtime without redeploying:

```
FEATURE_W1437=false
```

When disabled: - ticketSlaScheduler falls back to \$nin query -
nphiesReconciliationService.sweep falls back to pre-W1437 query - NphiesClaim pre-save
hook stops auto-updating updatedAt

Default is true.

10. Alert Integrations

The deploy scripts dispatch to:

- Slack (SLACK_WEBHOOK)
- Microsoft Teams (TEAMS_WEBHOOK)
- PagerDuty (PAGERDUTY_INTEGRATION_KEY)
- Sentry (SENTRY_DSN)

Loki alert rules route to AlertManager; configure AlertManager to forward to PagerDuty for page-worthy events.

11. Known Risks

Risk	Mitigation
Migration not run before deploy	deploy-w1437.sh fails fast; GitHub Actions workflow enforces order; --with-w1437-migration flag in deploy-vps.sh.
Indexes missing	Scripts verify indexes; Mongoose autoIndex builds them on startup if enabled.
Local changes lost on production host	--force flag required; script aborts on dirty working tree by default.
Rollback needed	rollback-w1437.sh automates code rollback and backup; GitHub Actions workflow available.
Long-running monitor forgotten	GitHub Actions workflow runs the full 30-minute monitor and reports result.

12. Sign-off

Role	Name	Sign-off	Date
Release owner			
DevOps / DBA			
On-call engineer			

13. Appendix: Required Compound Indexes

```
// AdvancedTicket
{ status: 1, 'sla.firstResponseAt': 1, 'sla.isBreached': 1, createdAt: -1 }

// NphiesClaim
{ 'nphies.submission.status': 1, 'nphies.submission.updatedAt': 1,
  'nphies.submission.submittedAt': 1 }
```

14. Appendix: Emergency Contacts

Add your team contacts here:

- On-call: _____
- DBA: _____
- Platform: _____

• Slack channel: _____